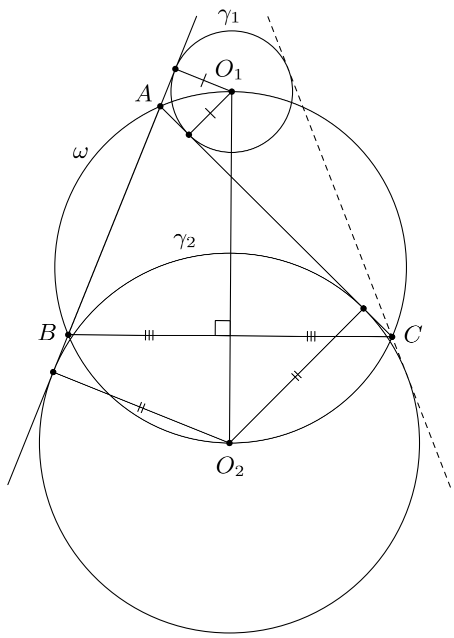
Problem 73. A triangle ABC with $AB < AC$ is inscribed in a circle ω. Circles γ_1 and γ_2 touch the lines AB and AC, and their centres lie on the circumference of ω. Prove that C lies on a common external tangent to γ_1 and γ_2.

Solution.



Let O_1 and O_2 be the circumcentres of γ_1 and γ_2, respectively. Note that the points O_1 and O_2 are at equal distances from the lines AB and AC, and so they lie on the external and internal bisectors of $\angle BAC$ (respectively in the figure above). Hence O_1 and O_2 are intersection points of external and internal bisectors of $\angle BAC$ and ω, that is, O_1 and O_2 are midpoints of $\widehat{BC}$ containing A and $\widehat{BC}$ not containing A (again respectively in the figure above). It follows that B and C are symmetric with respect to the line O_1O_2 passing through the centres of γ_1 and γ_2. Therefore if B lies on a common tangent to these circles, then so does C. $\square$

Problem 74. Find all positive integers $n \geqslant 2$ such that there exists a permutation $a_1, a_2, a_3, \ldots, a_{2n}$ of the numbers $1, 2, 3, \ldots, 2n$ satisfying

$$a_1a_2 + a_3a_4 + \ldots + a_{2n-3}a_{2n-2} = a_{2n-1}a_{2n}. \qquad (*)$$

Solution. The answers are $n = 3, 4, 5, 6$, and 7.

First observe that the minimum value of the left-hand side of $(*)$ is

$$\sum_{k=1}^{n-1} k(2n-1-k) = (2n-1)\cdot\frac{1}{2}(n-1)n - \frac{1}{6}(n-1)n(2n-1) = \frac{1}{3}n(n-1)(2n-1).$$

This is because for any two products ad and bc, we must have $a < b < c < d$ up to symmetry by the Rearrangement inequality, and we may assume the two numbers removed are $2n-1$ and $2n$. The above sum is the only case which has these properties.

Next we see (clearly) that the maximum value of the right-hand side of $(*)$ is $2n(2n-1)$ and so, together with the minimum value given above, we have

$$\frac{1}{3}n(n-1)(2n-1) \leqslant 2n(2n-1) \quad\Longrightarrow\quad n \leqslant 7.$$

Now we note that for $n = 2$, it impossible to satisfy the problem conditions. Lastly, we give examples for each of the answers above:

- for $n = 3$, we have $1\cdot 2 + 3\cdot 6 = 4\cdot 5$,
- for $n = 4$, we have $1\cdot 6 + 2\cdot 3 + 4\cdot 7 = 5\cdot 8$,
- for $n = 5$, we have $1\cdot 10 + 2\cdot 7 + 3\cdot 6 + 4\cdot 5 = 8\cdot 9$,
- for $n = 6$, we have $1\cdot 11 + 2\cdot 6 + 3\cdot 8 + 4\cdot 7 + 5\cdot 9 = 10\cdot 12$, and
- for $n = 7$, we have $1\cdot 12 + 2\cdot 11 + 3\cdot 10 + 4\cdot 9 + 5\cdot 8 + 6\cdot 7 = 13\cdot 14$. $\square$

Problem 75. A calculator can square a number or add 1 to it. It cannot add 1 two times in a row. After several operations it transformed a number x into a number $S > x^n + 1$ where x, n, S are positive integers.
Prove that $S \geqslant x^n + x - 1$.

Solution. We consider all the numbers modulo $x^2 + x + 1$. If the number in the calculator is congruent to x, then the number at the next step is congruent to x^2 or $x + 1 \equiv -x^2$. The number congruent to x^2 transforms either to $x^4 \equiv x$ or to $x^2 + 1 \equiv -x$. The numbers congruent to $-x^2$ and $-x$ are obtained by adding 1, so they can be only squared; this transforms them into numbers congruent to x and x^2, respectively. Thus we always get numbers congruent to $\pm x$ or $\pm x^2$ modulo $x^2 + x + 1$, or, equivalently, congruent to $x, x + 1, x^2, x^2 + 1$ modulo $x^2 + x + 1$.

Now note that if $n \equiv r \pmod 3$, then $x^n + 1 \equiv x^r + 1 \pmod{x^3 - 1}$ and so

$$x^n + 1 \equiv x^r + 1 \pmod{x^2 + x + 1}.$$

Thus $x^n + 1$ can leave only remainders $2, x + 1$ or $x^2 + 1$ when divided by $x^2 + x + 1$. On the other hand, the numbers in the calculator can leave only remainders $x, x + 1, x^2, x^2 + 1$. It is easy to check that if S (which gives one of the remainders $x, x + 1, x^2, x^2 + 1$) is greater than $x^n + 1$ (which gives one of the remainders $2, x + 1, x^2 + 1$), then the difference is at least $x - 2$. It follows that if $S > x^n + 1$, then

$$S \geqslant x^n + 1 + (x - 2) = x^n + x - 1. \qquad \square$$

Problem 76. Determine if there exist positive integers $a_1, a_2, \ldots, a_{10}, b_1, b_2, \ldots, b_{10}$ satisfying the following property: for each non-empty subset S of $\{1, 2, \ldots, 10\}$ the sum $\sum\limits_{i \in S} a_i$ divides the sum $12 + \sum\limits_{i \in S} b_i$.

Solution. The answer is no such numbers exist.

We first prove the following lemma:
Lemma: Among the integers a_1, a_2, a_3, a_4 and a_5 we can select some whose sum is divisible by 5.

Indeed, consider the following 5 integers: $a_1, a_1 + a_2, \ldots, a_1 + a_2 + a_3 + a_4 + a_5$. If they are all distinct modulo 5, there is a sum congruent to 0 modulo 5. Otherwise, by the Pigeonhole principle, there are two whose difference is congruent to 0 modulo 5, and the result follows.

Returning to the problem, we suppose positive integers $a_1, a_2, \ldots, a_{10}, b_1, b_2, \ldots, b_{10}$ exist satisfying the problem condition. Applying the lemma to the positive integers a_1, a_2, a_3, a_4, a_5, we get some a_i's whose sum is divisible by 5. If we denote the sum of the corresponding b_i's by n, then $n + 12$ is divisible by 5.

Applying lemma one more time to the positive integers $a_6, a_7, a_8, a_9, a_{10}$, we get some a_j's whose sum is divisible by 5. If we denote the sum of the corresponding b_j's by k, then $k + 12$ is divisible by 5.

Now observe that the sum of the a_i's and a_j's obtained above is also divisible by 5. The corresponding sum of the b_i's and b_j's is $n + k$. So $n + k + 12$ must be divisible by 5 and so

$$0 \equiv n + k + 12 = (n + 12) + (k + 12) - 12 \equiv 0 + 0 - 12 = -12 \pmod 5,$$

which is absurd. The result follows. $\square$